HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and debugged **System Verilog** testbenches and reusable **Universal Verification Components (UVCs)** for RISC-V microcontroller sub-systems, achieving **~85% functional coverage**.
- Implemented **constrained-random verification** strategies and directed **coverage-driven methodologies** to identify and resolve complex design bugs in embedded controller IPs.
- Assisted in integrating **RISC-V processor sub-systems** into higher-level SoC designs, contributing to test planning and debugging of test vector failures.
- Collaborated with design and validation teams to address and close **coverage gaps** identified during verification cycles, ensuring comprehensive testing.
- Wrote **Python scripts** for test automation, data analysis of simulation results, and generation of verification reports, reducing manual analysis time by **30%**.
- Participated in **Agile ceremonies**, including sprint planning and technical estimation, for the Microcontroller Sub-System IP team.
- Containerized verification environments using **Docker** for consistent execution across development and CI/CD pipelines.

SKILLS

Verification Methodologies – UVM, Formal Verification, Constrained-Random Verification, Coverage-Driven Methodologies

Languages & Scripting – System Verilog, Perl, Python, C/C++, Shell Scripting

Hardware Description & Simulation – RTL Simulation, Gate-Level Simulation, RISC-V ISA, ARM ISA

Tools & Protocols – UVM, Jasper, VC_Formal, AMBA Bus Protocols

Cloud & Containerization – AWS, Docker, Jenkins

PROJECTS

RISC-V Core Verification Environment

- Engineered a **System Verilog UVM testbench** for a custom RISC-V core, focusing on instruction set verification and pipeline integrity.
- Developed reusable **Universal Verification Components (UVCs)** for common interfaces and integrated **C models** for functional comparison.
- Achieved **90% code coverage** and **88% functional coverage** for the core instruction decode and execute stages.

Embedded Controller IP Integration

- Developed Python scripts to automate the generation and execution of test vectors for an **embedded controller IP**, targeting security and power management features.
- Collaborated with the SoC team to debug integration issues and ensure adherence to **AMBA bus protocols**, resolving 15+ critical bugs.
- Wrote preliminary runbooks for the verification of the IP within a larger system context.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)